DSHA-P49 Metal Multi-Factor Backtest Report

Report ID: 20261019_103453

Generated: 2026-10-19 10:34:53

Release Permit: ISSUED

Token: DSHA-P49-TOKEN (EXHAUSTED)

Data Source: wdf_sample_metals_long.csv

Symbols: CU.SHF, ZN.SHF, PB.SHF, AL.SHF, SN.SHF, NI.SHF

Date Range: 2025-01-02 ~ 2026-10-19

Strategy: multi_factor_composite_momentum

Factors: momentum_20d, spread_cs, oi_momentum_7d

Total Return (avg): -37.69%

Sharpe Ratio (avg): -1.2980

Max Drawdown (avg): -42.10%

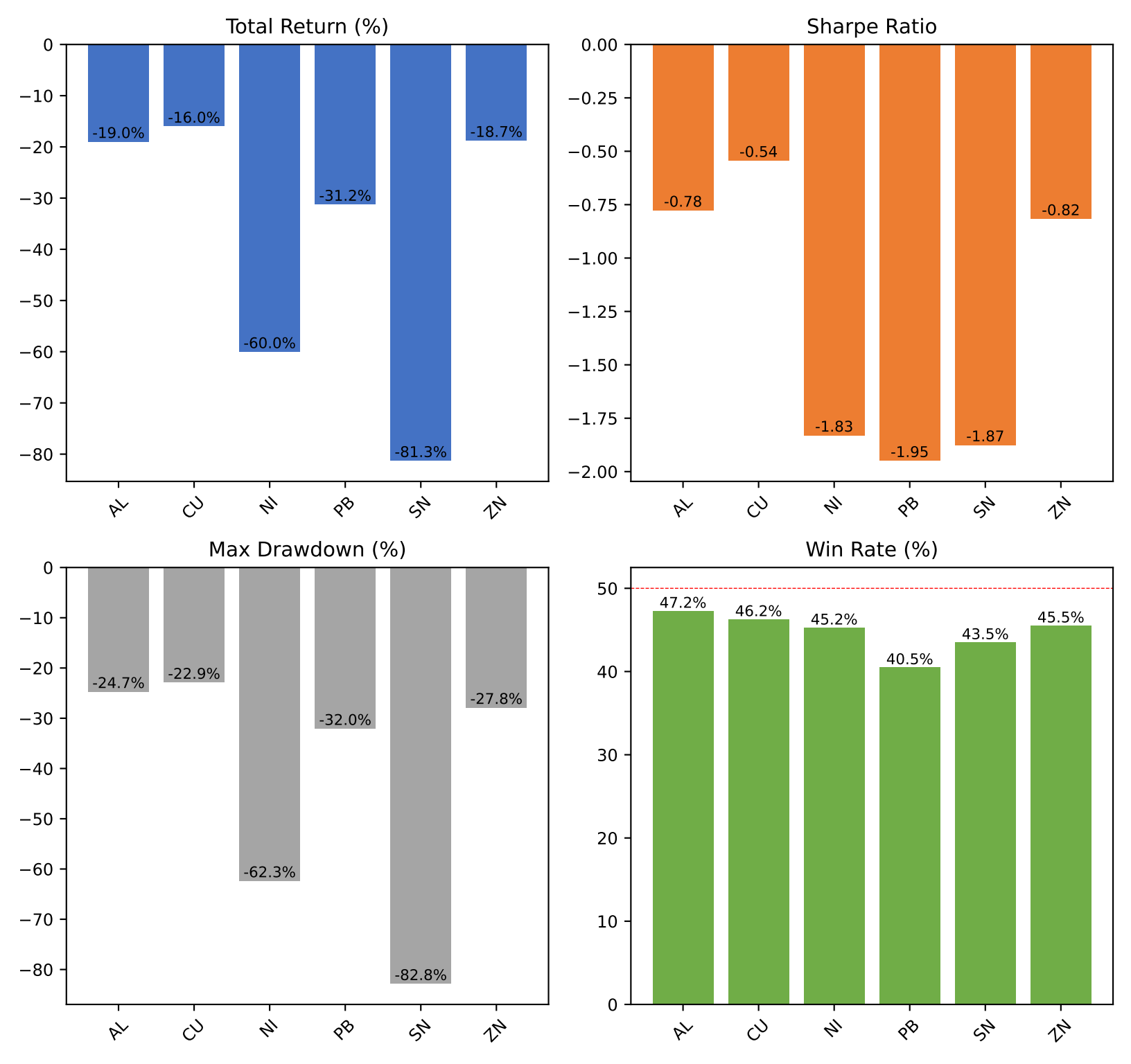
Win Rate (avg): 44.7%

REAL_WIND_CALL=False | WIND_API_CALLS=0

OFFLINE_ONLY=True | CSV_NOT_MODIFIED=True

SHFE 6-Metal Performance Comparison

Symbol	Return%	Annual%	Sharpe	MaxDD%	WinRate%	P/L	Trades
AL	-18.95	-12.46	-0.7761	-24.69	47.2	0.94	78
CU	-15.96	-10.42	-0.5431	-22.85	46.2	1.05	92
NI	-59.99	-44.01	-1.8294	-62.34	45.2	0.78	107
PB	-31.21	-21.09	-1.9484	-32.04	40.5	0.97	142
SN	-81.26	-65.37	-1.8748	-82.81	43.5	0.74	109
ZN	-18.74	-12.31	-0.8159	-27.85	45.5	1.03	114


Total Return (%)
0
−10
−20
−30
−40
−50
−60
−70
−80
-19.0%
-16.0%
-60.0%
-31.2%
-81.3%
-18.7%
AL
CU
NI
PB
SN
ZN
Sharpe Ratio
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
-0.78
-0.54
-1.83
-1.95
-1.87
-0.82
AL
CU
NI
PB
SN
ZN
Max Drawdown (%)
0
−10
−20
−30
−40
−50
−60
−70
−80
-24.7%
-22.9%
-62.3%
-32.0%
-82.8%
-27.8%
AL
CU
NI
PB
SN
ZN
Win Rate (%)
50
40
30
20
10
0
47.2%
46.2%
45.2%
40.5%
43.5%
45.5%
AL
CU
NI
PB
SN
ZN

Risk Metrics Summary

Portfolio Total Return (avg): -37.6868%
Portfolio Total Return (median): -25.0829%
Portfolio Annual Return (avg): -27.6112%
Portfolio Annual Vol (avg): 20.0146%
Portfolio Max DD (avg): -42.0961%
Portfolio Max DD (worst): -82.8087%
Portfolio Sharpe (avg): -1.2980
Portfolio Sharpe (median): -1.3227
Portfolio Win Rate (avg): 44.68%
Portfolio P/L Ratio (avg): 0.9193

Per-Symbol Detail:
  AL.SHF:
    Return: -18.95% | Annual: -12.46%
    Volatility: 16.05% | Sharpe: -0.7761
    Drawdown: -24.69% | Calmar: -0.5047
    Win Rate: 47.2% | P/L: 0.94
    Trades: 78 | Initial: 1,000,000
    Final: 810,458.14
  CU.SHF:
    Return: -15.96% | Annual: -10.42%
    Volatility: 19.20% | Sharpe: -0.5431
    Drawdown: -22.85% | Calmar: -0.4561
    Win Rate: 46.2% | P/L: 1.05
    Trades: 92 | Initial: 1,000,000
    Final: 840,410.24
  NI.SHF:
    Return: -59.99% | Annual: -44.01%
    Volatility: 24.06% | Sharpe: -1.8294
    Drawdown: -62.34% | Calmar: -0.7060
    Win Rate: 45.2% | P/L: 0.78
    Trades: 107 | Initial: 1,000,000
    Final: 400,059.11
  PB.SHF:
    Return: -31.21% | Annual: -21.09%
    Volatility: 10.83% | Sharpe: -1.9484
    Drawdown: -32.04% | Calmar: -0.6584
    Win Rate: 40.5% | P/L: 0.97
    Trades: 142 | Initial: 1,000,000
    Final: 687,883.51
  SN.SHF:
    Return: -81.26% | Annual: -65.37%
    Volatility: 34.86% | Sharpe: -1.8748
    Drawdown: -82.81% | Calmar: -0.7893
    Win Rate: 43.5% | P/L: 0.74
    Trades: 109 | Initial: 1,000,000
    Final: 187,380.58
  ZN.SHF:
    Return: -18.74% | Annual: -12.31%
    Volatility: 15.09% | Sharpe: -0.8159
    Drawdown: -27.85% | Calmar: -0.4421
    Win Rate: 45.5% | P/L: 1.03
    Trades: 114 | Initial: 1,000,000
    Final: 812,601.01

Safety & Compliance Statement

REAL_WIND_CALL = False
WIND_API_CALLS = 0
OFFLINE_ONLY = True
CSV_NOT_MODIFIED = True
P68_PARAMS_NOT_MODIFIED = True
DSHD_P81_NOT_INTERRUPTED = True
RELEASE_PERMIT = ISSUED
GATE = READY

Red Line Compliance:
  [X] DSHC original data NOT modified
  [X] Fully offline, zero Wind API calls
  [X] P68 parameter baseline NOT changed
  [X] DSHD-P81 audit observation NOT interrupted

Source CSV SHA256: 3c14f50f063c4734950064b1072585626588e93568191d6a85767ba1699b5ed1
Source CSV Size: 285191B

This report is generated from offline fixed CSV data.
No real-time Wind API calls were made.